

KallosumAI: Building a Clinical AI Product, and Proving Whether to Trust It

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The problem: the right answer, three days too late

I trained as a physician (MD in Pharmacology), and like many on that path, I was drawn toward Medical Affairs, the part of pharma that sits closest to the science while working directly with doctors. Before I ever joined the industry, I spent months talking to senior Medical Science Liaisons (MSLs) to understand the role. One thing came up in almost every conversation: a large part of an MSL's work is answering doctors' clinical questions, and that process is broken in a specific way.

A doctor raises a clinical question during an interaction with an MSL. The MSL takes it away, finds the evidence, routes it through the company's medical-information approval process, and returns with an answer one to three days later. By then the answer has often lost its relevance to the decision that prompted it.

That was the problem I set out to solve: get a trusted, evidence-backed answer to the doctor at the point of care, not days later.

The hypothesis

I had been following AI closely. I had even presented award-winning work on AI in antibiotic discovery during my MD, and I saw an opening. If a doctor's question could be answered instantly from a highly curated medical knowledge source, the answer would be there exactly when it was needed. Not a general chatbot drawing from the open web, but a focused assistant grounded in trusted evidence.

The MVP: learning to build by building

I built Version 1 in Flowise, a visual tool for wiring together AI components, and hosted it locally. It was simple, but building it taught me the whole stack from the ground up: how language models behave, how to write system prompts, how to build RAG workflows over a literature corpus, how to connect data stores, tools, and APIs, how to choose models, and how API keys and MCP servers work. Everything I built later rested on that foundation.

Then came the real question: would a doctor actually find this useful?

Testing it with practising doctors

I started with my family physician. I sat her down, showed her the assistant, and let her use it herself. She found it genuinely useful, but two pieces of her feedback shaped everything that followed. First, she pointed out something I had not considered: doctors are often wary of information coming straight from pharmaceutical reps, who aren't part of the medical fraternity,

whereas a tool built by another doctor carries real credibility. Second, she asked for one thing above all: a source citation on every answer.

She referred me to her husband, a paediatrician at the same hospital, and his feedback became one of the most valuable additions to the product. He asked for voice input, since a busy paediatrician rarely has time to type a long question. That single suggestion changed how the product worked.

From there I widened the circle: an orthopaedic surgeon, an ophthalmologist, a nephrologist, former classmates, and doctors I had trained under. Each brought a different angle, and slowly the shape of a genuinely useful product emerged. I folded all of it into Version 2, rebuilt with modern tooling, and paid attention to what makes tools like ChatGPT pleasant to use, adding response voting, copy and export, follow-up questions, and a cleaner interface. In its current form, the assistant brings evidence retrieval down from one to three days to under thirty seconds. You can try the assistant live at kallosumai.com/chat.

Getting inside the industry

At this point I had a product a network of doctors genuinely valued. But I was building from the outside, inferring how medical affairs actually worked rather than knowing it. So I made a deliberate move: I joined Bristol Myers Squibb as a Medical Science Liaison in hematology-oncology.

Once I was in, I learned far more than I had expected. Being on the field, having real scientific conversations with doctors, did two things. It brought a harder question about my own product into focus, and it revealed a second problem I could not have seen from the outside.

The harder question: whether it could be trusted

The question that matters most for a clinical tool is not whether doctors liked it, but whether its answers could be trusted. A confident, well-formatted, wrong answer is worse than no answer at all. So I stopped adding features and evaluated the product rigorously, looking specifically for where it failed.

I built a structured evaluation: 35 adversarial questions about a single drug, epoetin alfa, across five dimensions (factual grounding, clinical safety, completeness, source credibility, and latency), each scored Pass or Fail against a locked expert answer, with every failure coded into a taxonomy of recurring mechanisms. I then built two automated evaluators to scale the process, including an LLM-as-Judge validated against my own labels with a confusion matrix. The full method and findings are in [the technical report](#).

What the evaluation found

The evaluation surfaced a clear pattern. The failures were not random; they concentrated in the retrieval layer. The assistant often failed to reach its own curated corpus, or reached it but could not extract the specific answer, and it frequently leaned on non-authoritative web sources: 17 of

18 web-search answers cited at least one source such as a social-media page. There were also a few clinically serious misses, including a case where the assistant should have flagged a contraindication and did not.

None of this was visible from casual use. It surfaced only through systematic testing, which is the difference between a demo and a product that can be placed in front of a doctor.

The resulting roadmap

The evaluation did not just produce a score; it produced a prioritized engineering roadmap. The top fixes: make the assistant query its curated corpus before falling back to the web, restrict web sources to an approved authoritative list, and add explicit safety checks for high-risk clinical scenarios. Each fix is tied to specific failed cases and has a defined way to verify it in the next cycle. The evaluation became the instrument for determining whether the next version is genuinely better.

The second problem: what happens to everything doctors say

From the inside, I saw a second problem the assistant alone could not solve. Every day, MSLs have rich scientific conversations with doctors: their questions, concerns, sentiment, and the themes they return to again and again. Most of it is logged in generic CRM tools that were never built to capture the depth of a scientific exchange. The interaction is recorded, but the substance, the themes, the sentiment, and the strategic signal are largely lost.

So I built a second side to the platform: a perception mapper. It turns the assistant's chat queries and MSLs' post-visit notes into structured intelligence, tracking which clinical themes doctors ask about most, how sentiment toward a drug evolves visit over visit, where knowledge gaps cluster geographically, and what strategic actions should follow. It takes the unstructured output of field medical work and makes it analyzable.

On scope: the perception mapper is a working proof-of-concept, not something I have formally evaluated the way I evaluated the assistant. It reflects where I believe medical affairs is heading. The industry runs on these interactions, yet almost none of that signal is systematically captured today, and I expect that to change. You can explore the perception mapper at kallosumai.com/doctor-dashboard-2.

What colleagues say

KallosumAI has been reviewed by practising medical-affairs professionals, who spoke to its potential value for field teams.

"What impressed me most was the structured presentation of scientific information: concise summaries, comparison tables, and supporting references that simplify evidence review and support efficient, evidence-based scientific communication."

Dr. Rajasree G, Medical Science Liaison, Dr. Reddy's